

Recovering and Analysing Deleted Files on Windows

Project 5 • Recuva, FTK Imager evidence and Autopsy

Controlled recovery • forensic preservation • content and timeline analysis

Deleted data can remain evidential after a user removes it

Investigation objective

Recover deleted files, preserve disk evidence and interpret the remaining file-system artefacts in a repeatable Windows laboratory workflow.

- Controlled test files established known recovery targets.
- Recuva tested logical recovery from the Windows Recycle Bin.
- An E01 image enabled read-only examination in Autopsy.
- Metadata and timeline activity added investigative context.

Project 5: Recovering and Analyzing Deleted Files on Windows Systems

Introduction

Deleted files can still contain valuable information for forensic investigations. Recovering and analyzing these files can provide critical evidence of user activities or malicious actions. This project will guide you through the process of recovering and analyzing deleted files on Windows systems using various forensic tools.

Lab Setup

To complete this project, you will need access to a Windows operating system. You can use a physical machine or set up a virtual machine using software like VirtualBox or VMware.

Pre-requisites

- Basic understanding of Windows OS
- Familiarity with command-line interface
- Administrative privileges on the Windows machine

Tools Installation

For this project, we will use the following tools:

1. **Recuva:** A tool for recovering deleted files.
2. **FTK Imager:** A forensic imaging tool.
3. **Autopsy:** A digital forensics platform and graphical interface to The Sleuth Kit.

Installing Recuva

1. Download Recuva from the [official website](#).
2. Run the installer and follow the on-screen instructions.

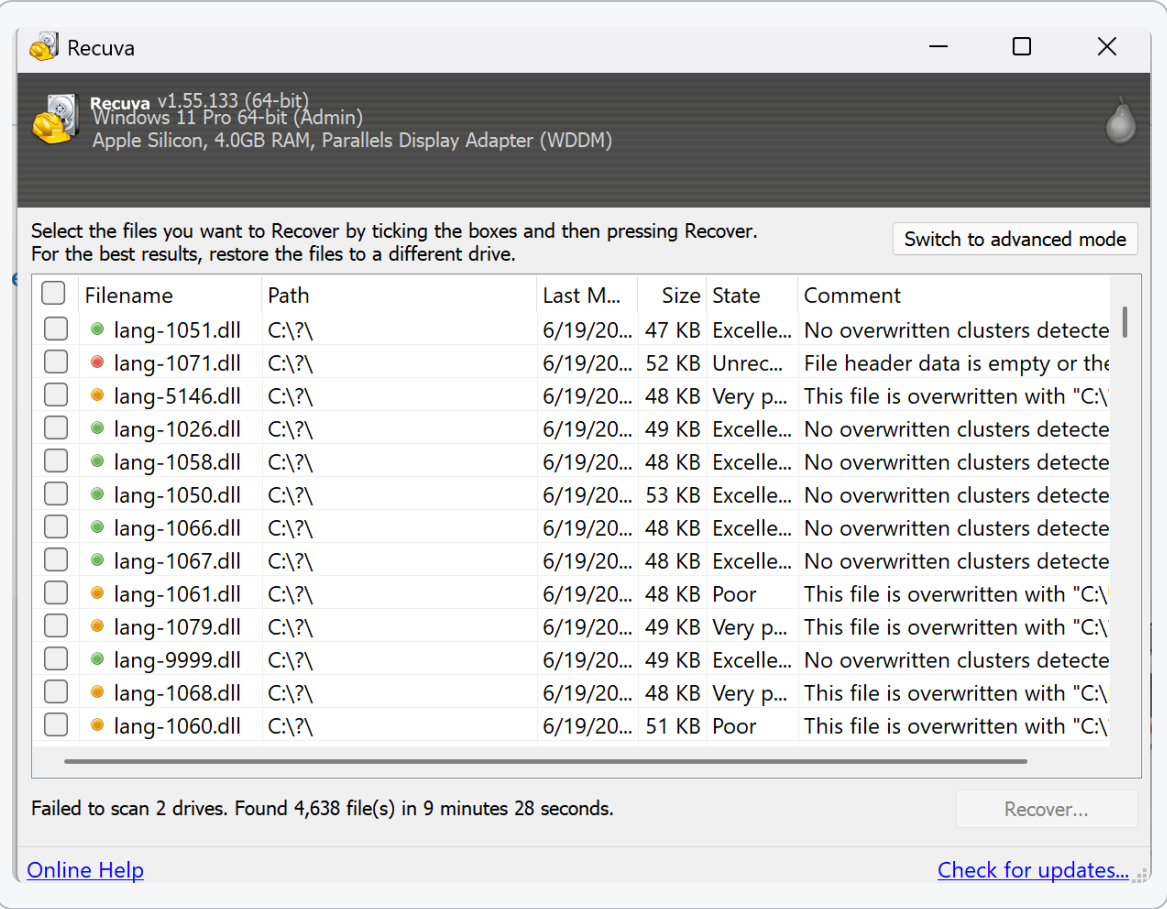
Installing FTK Imager

1. Download FTK Imager from the [AccessData website](#).
2. Run the installer and follow the on-screen instructions.

Installing Autopsy

1. Download Autopsy from the [official website](#).
2. Run the installer and follow the on-screen instructions.

Three tools covered recovery, preservation and interpretation



RECOVERY

Recuva 1.55.133

Located deleted records and reported recoverability based on overwritten clusters.

PRESERVATION

FTK Imager / E01

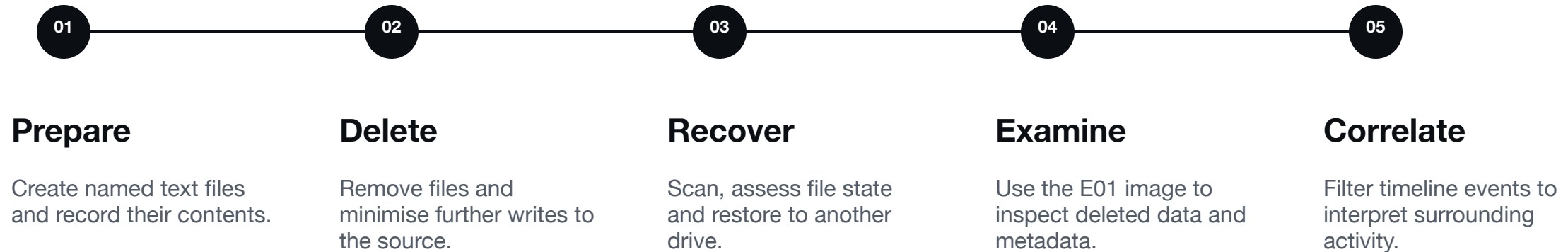
Provided a forensic container suitable for repeatable, read-only analysis.

ANALYSIS

Autopsy 4.23.1

Enumerated deleted records, displayed contents and correlated filesystem events.

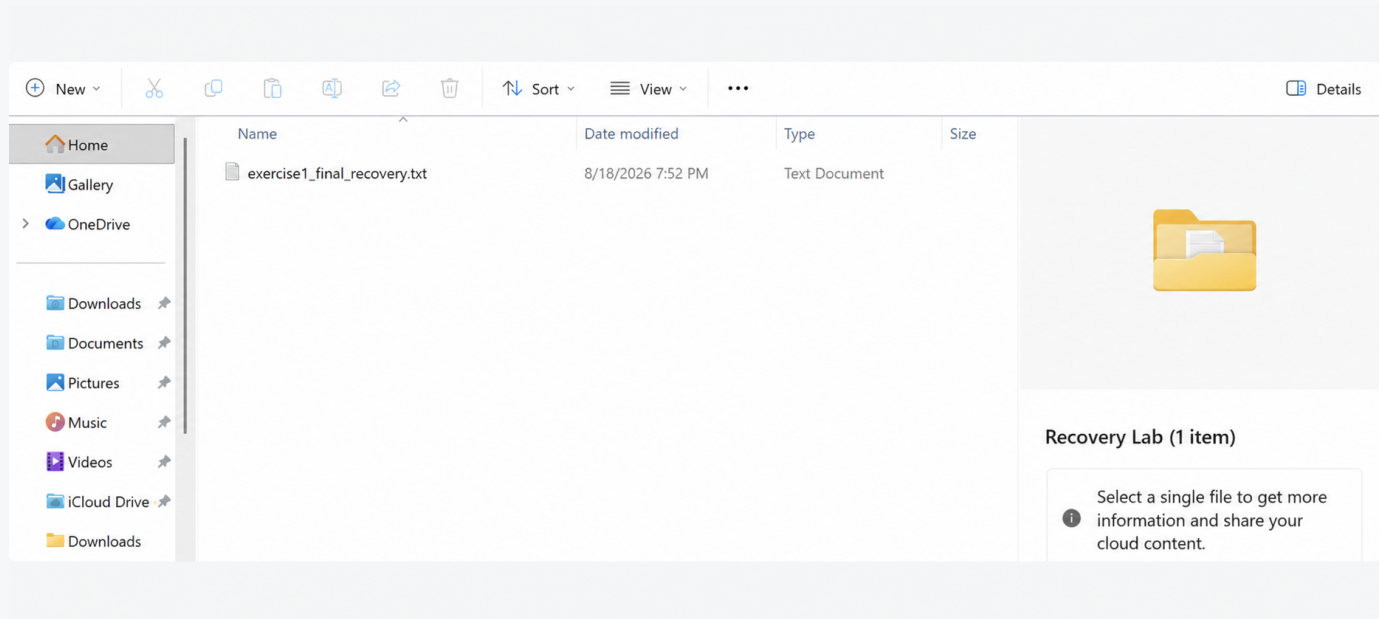
The workflow moved from a known test to contextual forensic analysis



Forensic controls

Preserve originals • recover to separate storage • document timestamps • state evidential limitations

A controlled dataset created a known recovery target



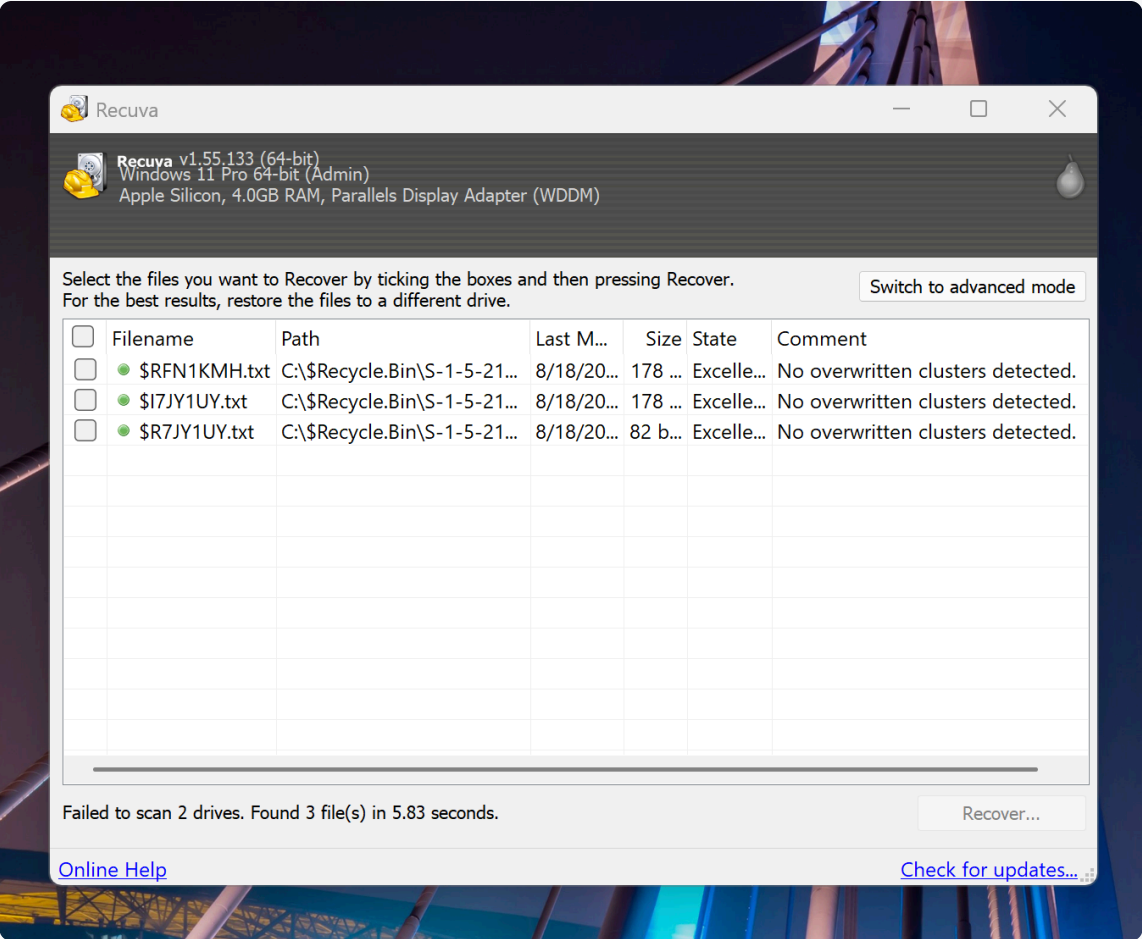
A uniquely named text document was created in Recovery Lab and populated with known content before deletion.

- exercise1_final_recovery.txt

Control point

Known filenames, sizes and contents made it possible to validate later recovery rather than merely locate anonymous deleted records.

Recuva found the deleted payload in an excellent recovery state



SELECTED EVIDENCE

\$R7JY1UY.txt

82 B

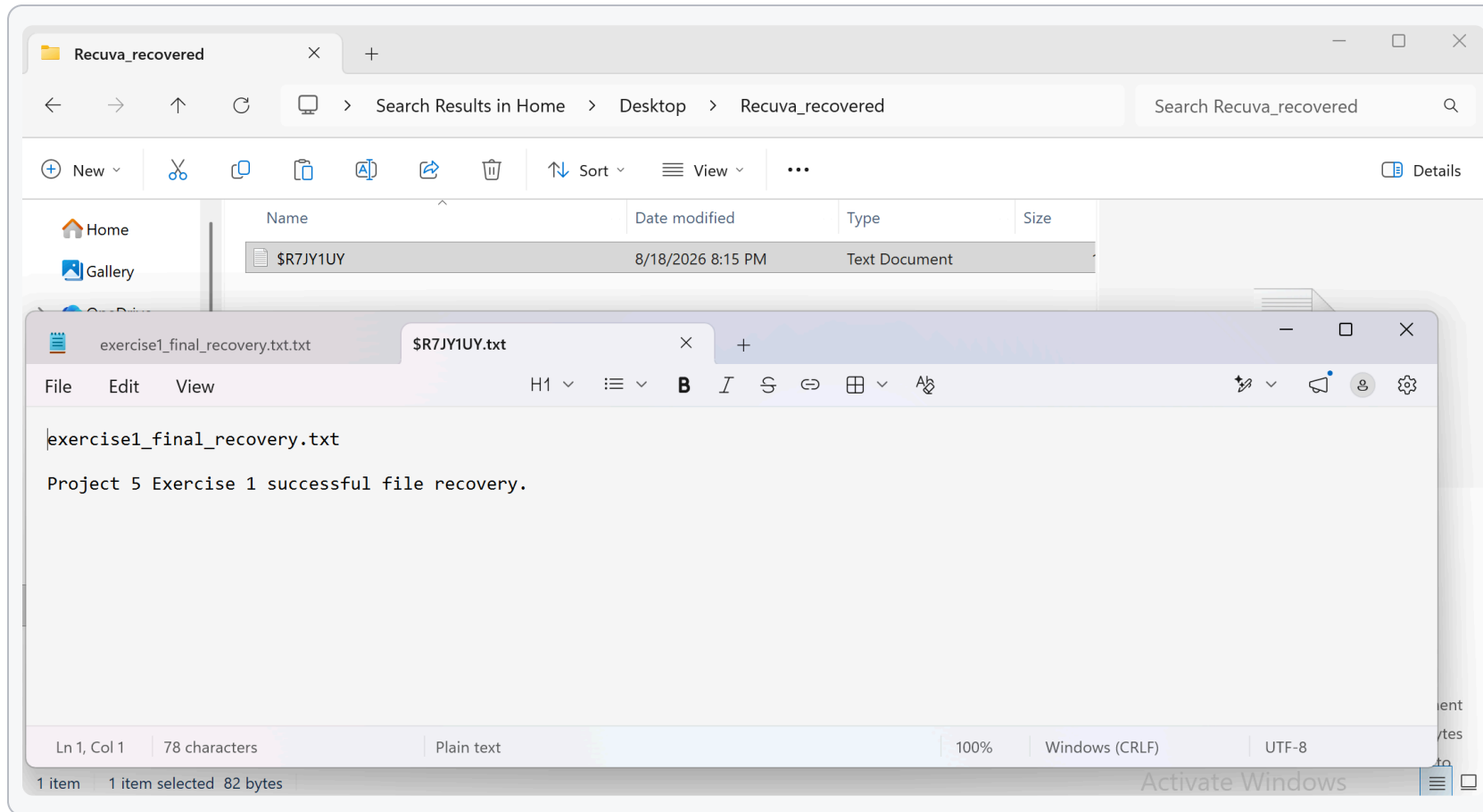
Recovered payload size

Excellent

No overwritten clusters detected

Recuva displayed the paired \$I7JY1UY.txt metadata record and \$R7JY1UY.txt payload. Another unrelated deleted record was present in the results but was not part of the controlled test.

The recovered file contained the expected controlled message



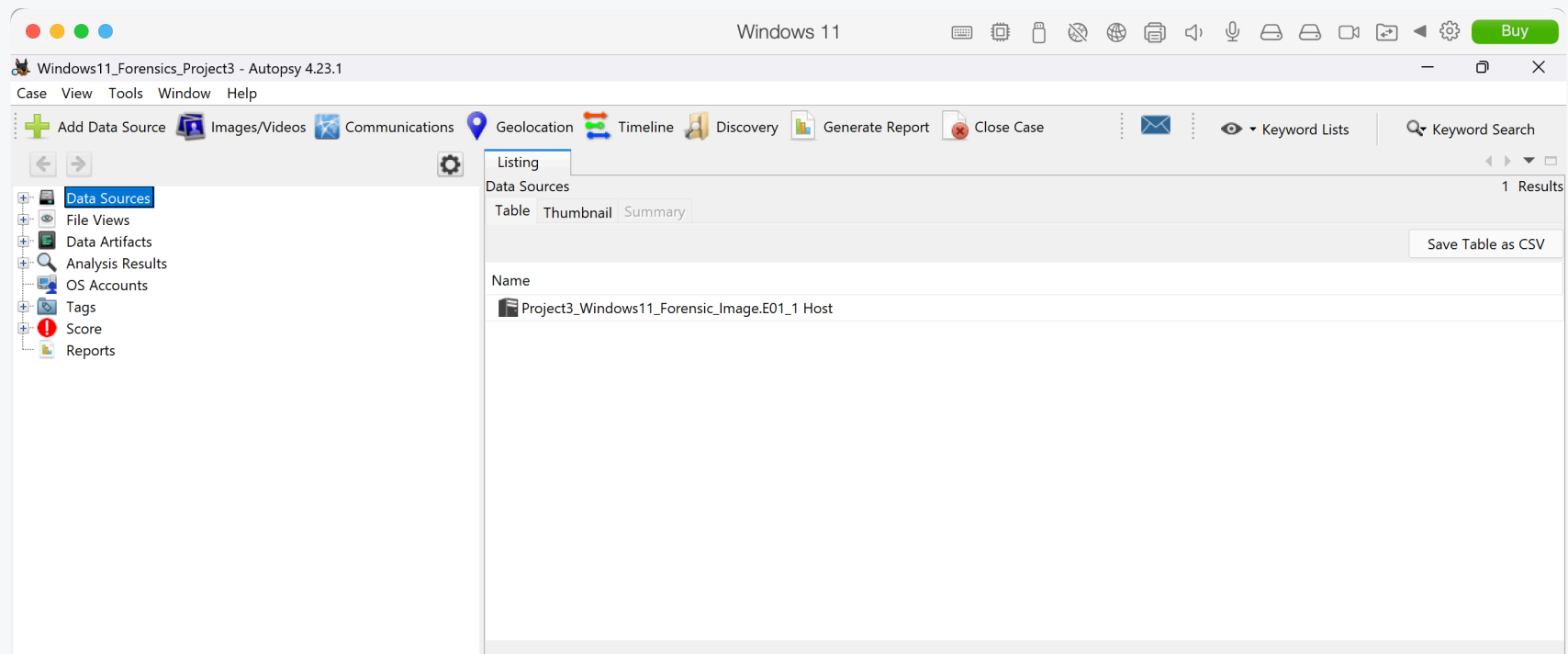
Recovery was confirmed by opening the restored payload and observing the expected filename and recovery message. Because the original pre-deletion content screenshot was unavailable, an independent byte-for-byte comparison was not documented.

OBSERVED CONTENT

“Project 5 Exercise 1 successful file recovery.”

Recovered to separate storage

An existing E01 image provided a stable source for Autopsy



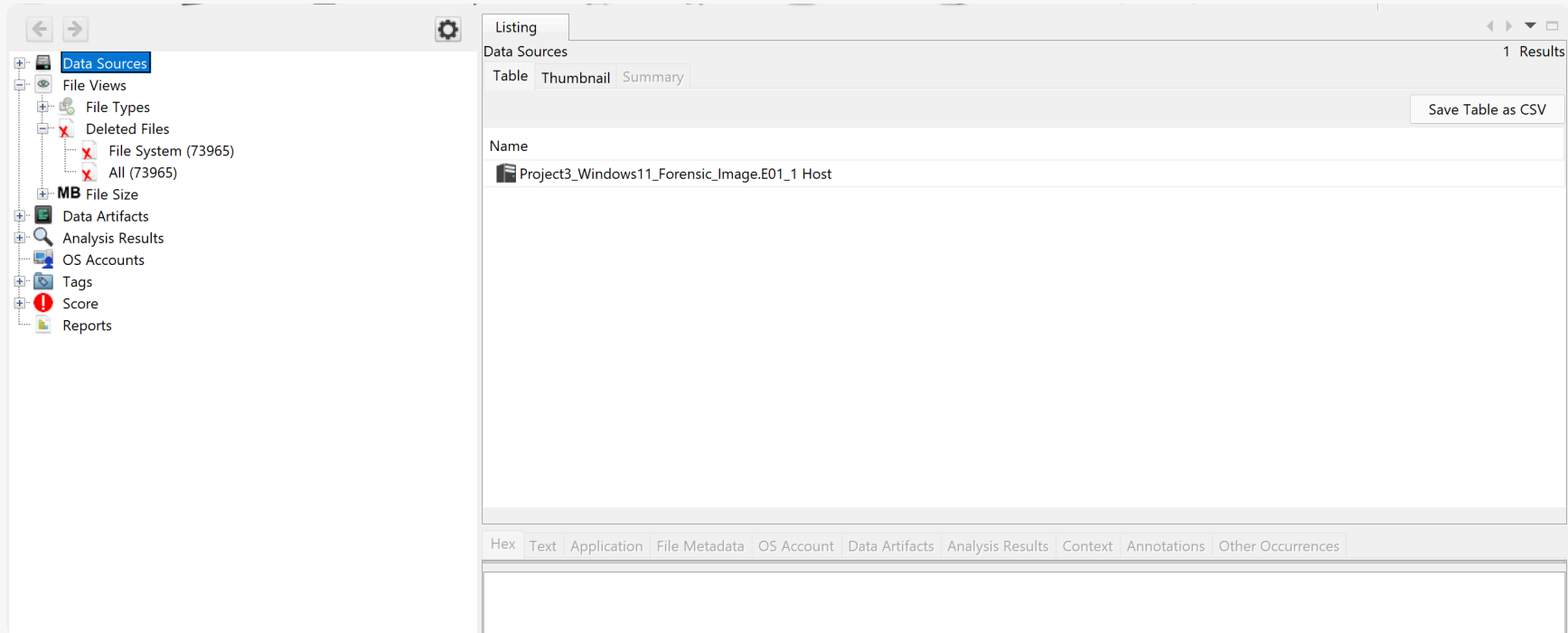
Evidence available

Project3_Windows11_Forensic_Image.E01 was successfully ingested as an Autopsy data source.

Documentation limitation

The acquisition and hash-verification dialogue was not captured in this screenshot set; ingestion confirms usability, not acquisition provenance by itself.

Autopsy enumerated 73,965 deleted file-system records



73,965

Deleted file-system records identified

The result demonstrates the scale of recoverable artefacts inside the image. Enumeration alone is not interpretation, so individual records were then examined for content and metadata.

A deleted Code Integrity policy remained inspectable at byte level

Selected artefact

{82443e1e-8a39-4b4a-96a8-f40ddc00b9f3}.cip

- Red X identified the record as deleted.
- Hex view exposed recoverable binary content.
- MIME context indicated a signed Code Integrity policy rather than a user document.

Interpretation Deleted does not mean content is immediately gone.

Listing

All

10000 Result:

Table Thumbnail Summary

Page: 1 of 8 Pages: Go to Page: Save Table as CSV

Name	S	C	O	Modified Time	Change Time
_ootaa64.efi				2026-08-12 13:03:48 BST	0000-00-00 00:00:00
bootaa64.efi.{7dcee70-c568-41eb-bbba-879a6f51ea03}				2026-08-12 13:03:48 BST	0000-00-00 00:00:00
{5DAC656C-21AD-4A02-AB49-649917162E70}.cip				2026-08-12 13:03:48 BST	0000-00-00 00:00:00
{82443e1e-8a39-4b4a-96a8-f40ddc00b9f3}.cip				2026-08-12 13:03:48 BST	0000-00-00 00:00:00
{CDD5CB55-DB68-4D71-AA38-3DF2B6473A52}.cip				2026-08-12 13:03:48 BST	0000-00-00 00:00:00
{5DAC656C-21AD-4A02-AB49-649917162E70}.cip.{877a472f-6113-42c1-85...				2026-08-12 13:03:48 BST	0000-00-00 00:00:00
{82443e1e-8a39-4b4a-96a8-f40ddc00b9f3}.cip.{c835a4e3-dfc1-4511-9613...				2026-08-12 13:03:48 BST	0000-00-00 00:00:00
{CDD5CB55-DB68-4D71-AA38-3DF2B6473A52}.cip.{f046db97-c6b9-41f5-a...				2026-08-12 13:03:48 BST	0000-00-00 00:00:00
boot.stl.{7ef08abd-0f6d-4402-8642-b97ce3936f74}				2026-08-12 13:03:48 BST	0000-00-00 00:00:00
D 02 ~3 D11				2026-08-12 13:03:48 BST	0000-00-00 00:00:00

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Page: 1 of 2 Page Go to Page: 1 Jump to Offset Launch in HxD

0x00000000: 30 82 7E F8 06 09 2A 86 48 86 F7 0D 01 07 02 A0 0.~...*.H.....
0x00000010: 82 7E E9 30 82 7E E5 02 01 01 31 0F 30 0D 06 09 ~.0.~...1.0...
0x00000020: 60 86 48 01 65 03 04 02 01 05 00 30 82 71 D7 06 ~.H.e.....0.q..
0x00000030: 09 2B 06 01 04 01 82 37 4F 01 A0 82 71 C8 04 82 .+.....70...q..
0x00000040: 71 C4 07 00 00 00 1E 3E 44 82 39 8A 4A 4B 96 A8 q.....>D.9.JK..
0x00000050: F4 0D DC 00 B9 F3 E4 F7 07 2E 4C 19 20 4D B7 C9L. M..
0x00000060: 6F 44 A6 C5 A2 34 84 25 10 B2 09 00 00 00 A3 00 oD...4.%.....
0x00000070: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Metadata established allocation state and recorded timestamps

Listing

All

TableThumbnailSummary

Page: 1 of 8Pages: < >Go to Page:

Name	S	C	O	Modified Ti
_ootaa64.efi				2026-08-12
bootaa64.efi.{7dcea70-c568-41eb-bbba-879a6f51ea03}				2026-08-12
{5DAC656C-21AD-4A02-AB49-649917162E70}.cip				2026-08-12
{82443e1e-8a39-4b4a-96a8-f40ddc00b9f3}.cip				2026-08-12
{CDD5CB55-DB68-4D71-AA38-3DF2B6473A52}.cip				2026-08-12
{5DAC656C-21AD-4A02-AB49-649917162E70}.cip.{877a472f-6113-42c1-85				2026-08-12
{82443e1e-8a39-4b4a-96a8-f40ddc00b9f3}.cip.{c835a4e3-dfc1-4511-9613				2026-08-12
{CDD5CB55-DB68-4D71-AA38-3DF2B6473A52}.cip.{f046db97-c6b9-41f5-a				2026-08-12
boot.stl.{7ef08abd-0f6d-4402-8642-b97ce3936f74}				2026-08-12
_D_02_~3.DLL				2026-08-12
kd_02_14e4.dll				2026-08-12

HexTextApplicationFile MetadataOS AccountData ArtifactsAnalysis ResultsContextAnnotation

Size:31409

File Name Allocation:Unallocated

Metadata Allocation:Unallocated

Modified:2026-08-12 13:03:48 BST

Accessed:2026-08-12 00:00:00 BST

Created:2026-08-12 13:04:25 BST

Changed:0000-00-00 00:00:00

FILE STATE

Unallocated

Both filename and metadata allocation were marked unallocated.

31,409 bytes

Modified12 Aug 2026 · 13:03:48 BST

Created12 Aug 2026 · 13:04:25 BST

Accessed12 Aug 2026 · 00:00:00 BST

The zero 'Changed' value indicates unavailable metadata, not a failed examination.

Timeline correlation linked the file to grouped Code Integrity activity

Display Times In: ☒ Local Time Zone ☐ GMT / UTC

History ← Back Forward →

Filters

Apply Reset

☒ Must include text: CIPolicies

☐ Must be tagged

☐ Must have hash hit

☐ Limit data sources to
Project3Windows11_Fore...

☐ Limit file types to
Media
Documents
Executables
Other

☐ Limit event types to
File System
File Accessed
File Changed

Hidden Descriptions

View Mode: Counts Details List

+ Add Event Snapshot Report Refresh View

Date/Time	Event Type	Description	Tagged	Hash Hit
2024-04-01 09:43:02	_BM	/Windows/System32/CodeIntegrity/CIPolicies/Active/{0283AC0F-FFF1-49AE-ADA1-8A933130CAD6}.cip		
2024-04-01 09:43:19	_BM	/Windows/System32/CodeIntegrity/CIPolicies/Active/{0939ED82-BFD5-4D32-B58E-D31D3C49715A}.cip		
2024-04-01 09:43:19	_BM	/Windows/Boot/EFI/CIPolicies/Active/{5DAC656C-21AD-4A02-AB49-649917162E70}.cip		
2024-04-01 09:43:20	_M	/EFI/Microsoft/Boot/CIPolicies/Active/{5DAC656C-21AD-4A02-AB49-649917162E70}.cip		
2024-04-01 09:43:24	_BM	/Windows/Boot/EFI/CIPolicies/Active/{CDD5CB55-DB68-4D71-AA38-3DF2B6473A52}.cip		
2024-04-01 09:43:26	_M	/EFI/Microsoft/Boot/CIPolicies/Active/{CDD5CB55-DB68-4D71-AA38-3DF2B6473A52}.cip		
2024-04-01 09:43:26	_BM	/Windows/System32/CodeIntegrity/CIPolicies/Active/{1283AC0F-FFF1-49AE-ADA1-8A933130CAD6}.cip		
2024-04-01 09:43:27	_BM	/Windows/System32/CodeIntegrity/CIPolicies/Active/{1939ED82-BFD5-4D32-B58E-D31D3C49715A}.cip		
2024-04-01 09:45:31	_B_	/Windows/System32/CodeIntegrity/CIPolicies/Active		
2024-04-01 09:45:31	_B_	/Windows/Boot/EFI/CIPolicies/Active		
2024-04-01 09:45:31	_B_	/Windows/System32/CodeIntegrity/CIPolicies		
2024-04-01 09:45:31	_BM	/Windows/Boot/EFI/CIPolicies		
2024-04-01 09:45:31	_BM	/Windows/System32/CodeIntegrity/CIPolicies/Staged		
2024-04-01 09:45:33	_B_	/Windows/WinSxS/FileMaps/\$\$_system32_codeintegrity_cipolicies_staged_276d48e6ef8844ae.cdf-ms		
2024-04-01 09:45:33	_B_	/Windows/WinSxS/FileMaps/\$\$_system32_codeintegrity_cipolicies_active_29d3e16aea6e0340.cdf-ms		
2024-04-01 09:45:34	_B_	/Windows/WinSxS/FileMaps/\$\$_boot_efi_cipolicies_active_8937710587dbb730.cdf-ms		
2026-07-11 20:48:41	_BM	/Windows/System32/CodeIntegrity/CIPolicies/Active/{15EA545D6-2A58-42E5-90A2-DD9E85681D3C}.cip		

Start: May 30, 2007, 12:21:54 AM

Jump By: year

End: Aug 13, 2026, 12:00:00 AM

Activate Windows

74

Timeline events matched the CIPolicies filter

Related paths appeared across Windows System32, Windows Boot EFI and EFI Microsoft Boot. Closely grouped timestamps suggest automated policy deployment, update or servicing activity.

Inference—not proof
Confirm with Windows Update and event logs.

All five workflows were evidenced, with one documentation caveat

01	Recover deleted files with Recuva	Complete
02	Create / use a forensic E01 image	E01 ingested in Autopsy*
03	Analyse deleted files in Autopsy	Complete
04	Examine recovered-file metadata	Complete
05	Correlate files with timeline events	Complete

82 B

Controlled payload recovered intact

73,965

Deleted records enumerated

74

Related timeline events

*Acquisition/hash-verification screenshot was not available in this session.

Deletion removed references—not every trace of the underlying activity

What the investigation demonstrated

- Recovery likelihood depends on whether storage clusters have been overwritten.
- Recycle Bin \$R and \$I records separate payload data from deletion metadata.
- Content, metadata and timeline evidence are stronger when interpreted together.
- Binary artefacts can still be examined even when they are not human-readable.

Improvements for the next acquisition

- Capture FTK Imager source selection, image settings and verification hashes.
- Record timezone and system clock assumptions at the start of the case.
- Recover to dedicated external media and calculate hashes of restored files.
- Correlate with Windows event logs before assigning a cause to deletion.

Outcome: a defensible workflow recovered known data and added system-level context without overstating the evidence.